TaskFlow

Plan • Organize • Achieve

Risk Assessment & Mitigation Plan

Personal & Collaborative Project Management Platform

PROGRAM	**DEPI**
DOCUMENT	**Risk Assessment & Mitigation Plan**
VERSION	**1.0**
PREPARED	**14 October 2026**
TEAM SIZE	**5 members**
AI	**Out of scope**

Scope note: TaskFlow is intended for individuals, freelancers, teams, and companies. The plan covers the authenticated application and the agreed features: workspaces, projects, Kanban, task management, time tracking, client portal/Project Live Room, deliverable review, change requests, handover notes, reporting, and rule-based project-health alerts. Public marketing pages are handled separately by Mostafa and are excluded from the shared module risk ownership below.

1. Purpose

This document identifies risks that could affect the delivery, security, reliability, usability, or quality of TaskFlow. It defines preventive measures, contingency actions, and an accountable owner for each risk. The risk register is a working document and must be reviewed as implementation progresses.

2. Risk Rating Method

Each risk is rated from 1 to 5 for likelihood and impact.

- **Likelihood:** 1 = Rare, 2 = Unlikely, 3 = Possible, 4 = Likely, 5 = Very likely.
- **Impact:** 1 = Negligible, 2 = Minor, 3 = Moderate, 4 = Major, 5 = Critical.
- **Risk Score:** Likelihood × Impact.
- **Priority:** 15–25 = High, 8–14 = Medium, 1–7 = Low.

These are initial qualitative estimates for planning, not measured probabilities. The team should update the ratings based on actual progress, test results, and decisions.

3. Risk Register

ID	Risk	Likelihood	Impact	Score / Priority	Preventive Mitigation	Contingency / Response	Accountable Owner
R-01	**Scope creep:** The feature set becomes too large for the DEPI schedule.	5	5	25 High	Define a minimum viable release; tag features as Must-have or Optional; use acceptance criteria; approve scope changes before adding work.	Defer optional features such as advanced analytics, complex automation, billing, or advanced Gantt calculations; keep the core workflow stable.	Mostafa Mohamed Mahmoud Mohamed (Team Lead), with all members
R-02	**Schedule slippage:** Implementation, integration, testing, or documentation takes longer than planned.	4	5	20 High	Track tasks in GitHub Issues; review progress and blockers weekly; complete integration in small increments; keep testing and documentation active during implementation.	Reduce scope to the agreed demo path; focus on critical defects; document deferred items and limitations rather than presenting unfinished features as complete.	Mostafa (Team Lead); each module owner for their work
R-03	**Unauthorized access to projects or client information.**	3	5	15 High	Enforce server-side authorization for every relevant page, action, file, and query; test direct URL/API access; validate workspace/project/client relationships. Do not rely only on hiding UI controls.	Block release of affected workflows; fix access checks; repeat negative permission tests and review access logs before enabling the feature again.	Shared: Ahmed for workspace/project access, Eman for client access, Mostafa for auth conventions
R-04	**Timer inaccuracies or duplicate time entries** after browser refresh, interruption, repeated clicks, or a forgotten stop action.	3	4	12 Medium	Persist session state on the server; use server timestamps; define pause/resume semantics; decide whether one active timer per user is allowed; make start/stop operations safe against duplicate requests.	Provide an auditable correction process for incomplete sessions; mark questionable entries for review rather than silently changing reported time.	Ahd Ammar Eisa El-Baqary
R-05	**Database migration conflicts** when multiple members change shared entities or migrations concurrently.	4	4	16 High	Agree on entity ownership and schema contracts; communicate changes before committing migrations; coordinate migration integration through one designated integrator; avoid multiple simultaneous migrations against the same model.	Pause merges that affect the schema; reconcile the model and regenerate or repair migrations in a controlled branch; test upgrade from a clean database.	Mostafa coordinates; all entity owners contribute
R-06	**Module integration failure** caused by mismatched IDs, service contracts, status names, or data assumptions across members.	4	4	16 High	Agree on common entity names, DTOs, status enums, and service interfaces; use small pull requests; integrate early; document cross-module dependencies.	Create a focused integration fix branch; temporarily simplify the affected interaction while preserving data correctness; add regression tests before merging.	Mostafa coordinates; affected module owners resolve their parts
R-07	**SignalR updates are missing, delayed, or sent to the wrong users.**	3	4	12 Medium	Treat the database as the source of truth; authorize SignalR connections and groups; publish events after successful changes; handle reconnects and reload current server state.	Fall back to re-fetching server data or manual refresh; disable problematic real-time events until correct authorization and delivery are restored.	Mostafa for shared setup; each module owner for emitted events

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R-08	**Unsafe or inaccessible file attachments** due to invalid file types, excessive size, public file paths, or incorrect permissions.	3	4	12 Medium	Validate extensions, actual content/type, and size; keep files in controlled storage; authorize downloads; avoid trusting client-supplied MIME types; do not expose private files through guessable public URLs.	Disable affected upload paths; remove or quarantine unsafe files; repair access configuration and retest downloads using authorized and unauthorized accounts.	Alaa Essam Hamed Abdelkader
R-09	**Gantt/dependency logic becomes too complex** or presents misleading dates and critical-path information.	4	3	12 Medium	Limit the first version to explicitly supported dependency types; validate date ranges and cycles; test representative project schedules; distinguish supported calculations from visual-only features.	Deliver a simpler timeline view and defer advanced critical-path calculation until it can be validated.	Ahd Ammar Eisa El-Baqary
R-10	**Hosting or cloud-service constraints** prevent the planned deployment or cause unexpected costs.	3	4	12 Medium	Check Azure access, credits, quotas, and database/storage requirements early; keep configuration environment-specific; avoid paid services without approval.	Use an approved alternative host or local demonstration, document the environment and limitations, and preserve a repeatable setup guide.	Mostafa coordinates; all members support deployment fixes
R-11	**Insufficient testing before final submission.**	4	5	20 High	Write acceptance criteria with each feature; test during development; reserve time for regression; test critical paths across module boundaries, including client isolation and timer correctness.	Freeze features; fix all known critical/high-severity defects; publish a known-issues list for non-critical problems and avoid claiming untested functionality.	All members; Mostafa coordinates end-to-end testing
R-12	**Progress percentage is misleading** because tasks have different sizes or cancelled/archived tasks are counted incorrectly.	3	3	9 Medium	Document the progress formula; define treatment of archived, cancelled, weighted, and blocked tasks; test the dashboard against known sample data.	Label the metric clearly and correct the formula; show task counts alongside percentages if a weighted model is not ready.	Ahd for reports; Ahmed/Alaa for source data contracts
R-13	**Client review or change-request decisions are unclear or disputed.**	3	4	12 Medium	Save review decisions against a specific deliverable version with reviewer, timestamp, and feedback; separate a requested change from an approved change; define who may approve.	Escalate unclear decisions to the project manager; keep the previous approved version and record the dispute or revised decision.	Eman Mohamed Masad El-Metwally
R-14	**Third-party UI libraries** (drag-and-drop, Gantt, charts) have compatibility, accessibility, or maintenance issues.	3	3	9 Medium	Prototype selected libraries early; check browser behaviour and licences; isolate library-specific code behind small components; keep a simpler fallback for essential functions.	Replace the library or simplify the interaction if it blocks core task management; avoid allowing a chart issue to corrupt saved task data.	Alaa for Kanban; Ahd for Gantt/charts
R-15	**Notifications become noisy, duplicated, or irrelevant.**	3	3	9 Medium	Define event types and recipient rules; allow reasonable preferences; avoid sending multiple notifications for one logical event; distinguish in-app events from email digests.	Temporarily disable the noisy rule, correct event/recipient logic, and restore it after testing.	Mostafa for shared notification service; module owners for event definitions

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R-16	**Poor usability from an overly complex interface**, especially for individual users who do not need team/client features.	3	4	12 Medium	Make personal use the default starting path; make team and client workflows optional; test main journeys with a few representative users; use consistent navigation and clear empty states.	Simplify navigation and hide irrelevant actions based on permissions and context; defer secondary screens from the first release.	Mostafa for personal dashboard/navigation; all frontend owners
R-17	**Data loss or an unrecoverable mistake** affects project, task, time, or review history.	2	5	10 Medium	Use database migrations safely; test archive/restore and deletion rules; maintain backup/export guidance appropriate to the environment; protect audit and review history from accidental overwrites.	Restore from the latest available backup/export if possible; document the incident and validate restored data before resuming normal use.	Ahmed for project/workspace data; each owner for module data; Mostafa coordinates release/backup guidance
R-18	**Team availability or skill gaps** leave a critical module unfinished or dependent on one person.	3	4	12 Medium	Assign one owner and a documented backup reviewer; keep setup notes and pull requests visible; review workload and blockers; share knowledge on cross-module contracts.	Reallocate tasks, pair on the critical path, or reduce scope with team agreement.	Mostafa and all members
R-19	**Requirements or public-page work compete with internal application work.**	3	3	9 Medium	Keep the public marketing pages assigned separately to Mostafa; maintain explicit scope boundaries and milestone priorities; document which items count as internal application deliverables.	Defer optional marketing polish or non-critical internal enhancements while protecting DEPI-required deliverables and the working core application.	Mostafa (Team Lead)
R-20	**Sensitive settings or secrets are committed to GitHub.**	2	5	10 Medium	Use environment variables or a secure secret store; commit `.env.example` without actual credentials; review configuration files before pushing; restrict repository access as appropriate.	Rotate exposed credentials immediately, remove the secret from active configuration and repository history where practical, and review logs for misuse.	Mostafa coordinates; every contributor is responsible for their commits

4. Priority Response Plan

The risks below require explicit review at each milestone because their impact could prevent a secure or successful demonstration.

High-priority risks

- **R-01 Scope creep:** Maintain a small, demonstrable core release and defer optional features when needed.
- **R-02 Schedule slippage:** Review blockers weekly and integrate modules early.
- **R-03 Client/workspace data exposure:** Verify authorization at the server and test forbidden access paths.
- **R-05 Migration conflicts:** Coordinate schema changes and migration merges.
- **R-06 Integration failure:** Agree on interfaces and run cross-module tests continuously.
- **R-11 Insufficient testing:** Start testing with the first implemented feature; reserve the last implementation days for stabilization.

Incident handling

If a high-impact security or data-integrity problem is discovered:

1. Stop or disable the affected operation if it risks exposing or corrupting data.
2. Notify the team lead and the owner responsible for the affected module.
3. Reproduce the issue and identify its scope.
4. Fix the root cause and add a regression test.
5. Re-test both permitted and forbidden scenarios before re-enabling the feature.
6. Record the incident, action taken, and any remaining risk in the project log.

5. Risk Monitoring Process

- Review the register at least once per week and at every milestone review.
- Update likelihood, impact, score, owner, status, and next action when new evidence appears.
- Track mitigation work in GitHub Issues; high-priority risks should have a specific action owner and due date.
- Raise any threat to client privacy, saved time accuracy, core workflow integrity, or a DEPI milestone as soon as it is identified.
- Reassess risks after major design changes, integration, deployment, and user testing.

Suggested status values

`Open` · `Mitigation in progress` · `Monitoring` · `Closed` · `Accepted with reason`

6. Ownership by Team Member

Member	Risk responsibilities
Mostafa Mohamed Mahmoud Mohamed — Team Lead	Scope and schedule coordination, architecture/integration, shared authentication conventions, notification/SignalR foundation, secrets handling coordination, deployment and release readiness. Public marketing pages are a separate workload.
Ahmed Reda Ahmed Sedik	Workspace/project access isolation, memberships and invitations, project data correctness, and project-level archive/restore behaviour.
Alaa Essam Hamed Abdelkader	Task workflow integrity, Kanban ordering/status rules, comment/attachment permissions, handover records, and safe file upload/download behaviour.
Ahd Ammar Eisa El-Baqary	Timer lifecycle and duration accuracy, time reports, Gantt/dependency behaviour, and project-health alert rules.
Eman Mohamed Masad El-Metwally	Client data isolation, Project Live Room visibility, deliverable version/review history, approval decisions, and change-request lifecycle.

Risk ownership does not mean a risk can only be mitigated by one person. Cross-module risks require collaboration, and the team lead coordinates escalation and decision-making.

7. Residual Risk and Limitations

Even after mitigation, some risks may remain. In particular, the breadth of the proposed feature set and the limited time between the implementation deadline (30 November 2026) and final submission (4 December 2026) mean that prioritization is essential. Azure deployment also depends on account access and service limits. The team must report actual test coverage, deployment status, and unresolved defects honestly in the final report.

The ratings above are initial estimates and must be reviewed by the team. They are not statistical predictions or official DEPI grading criteria.

8. Approval

Reviewer	Name	Status / Date
Team Lead	Mostafa Mohamed Mahmoud Mohamed	Pending team review
Team Representative	To be confirmed	Pending
DEPI Instructor / Reviewer	To be confirmed	Pending